New Pricing Models, Same Old Phillips Curves?

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Motivation

Central question of monetary economics: the nature of nominal rigidities

- **"Old" time-dependent (TD) models** (Calvo, Taylor):
  tractable, give the New Keynesian Phillips curve

$$\pi_t = \kappa \cdot \widehat{mc}_t + \beta\, \mathbb{E}_t[\pi_{t+1}] \tag{NKPC}$$

- **"New" state-dependent (SD) models** (menu cost: Golosov–Lucas, Nakamura–Steinsson):
  match the rich micro evidence on price changes

Main result of the paper

an **exact first-order equivalence between SD and TD** - Phillips curve of the canonical menu cost model is the same as that of a mixture of two TD models.

numerical first-order equivalence - for a wide range of common parameterizations, this linearized Phillips curve is numerically almost identical to the Calvo Phillips curve (NK-PC), for some κ

Roadmap

1. The Generalized Phillips Curve
2. The Pass-Through Matrix
3. State-Dependent (Menu Cost) Models
4. Time-Dependent Models
5. Exact Equivalence: SD = mixture of 2 TD models
6. Numerical equivalence to Calvo

Definitions

Generalization of Phillips curve for any pricing model

Given an impulse response $\{\widehat{mc}_t\}_{t\geq 0}$ of real marginal costs, any price-setting model produces an impulse response $\{\pi_t\}_{t\geq 0}$ of inflation.

Definition (Generalized Phillips Curve)

The **generalized Phillips curve** $\mathcal{K}$ is the linear map from $(\widehat{mc}_0, \widehat{mc}_1, \dots)$ to $(\pi_0, \pi_1, \dots)$

$$\underbrace{\begin{pmatrix} \pi_0 \\ \pi_1 \\ \pi_2 \\ \vdots \end{pmatrix}}_{\boldsymbol{\pi}} = \mathcal{K} \cdot \underbrace{\begin{pmatrix} \widehat{mc}_0 \\ \widehat{mc}_1 \\ \widehat{mc}_2 \\ \vdots \end{pmatrix}}_{\widehat{\boldsymbol{mc}}}$$

i.e. the sequence-space Jacobian

The Calvo model as a special case

For the standard Calvo model with parameter λ, we can solve $\pi_t = \kappa \cdot \widehat{mc}_t + \beta\,\mathbb{E}_t[\pi_{t+1}]$ forward:

$$\pi_t = \kappa \sum_{j=0}^{\infty} \beta^j \widehat{mc}_{t+j}$$

$$\mathcal{K} = \begin{pmatrix} \kappa & \beta\kappa & \beta^2\kappa & \cdots \\ 0 & \kappa & \beta\kappa & \cdots \\ 0 & 0 & \kappa & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \qquad \kappa = \frac{\lambda\,(1-\beta(1-\lambda))}{1-\lambda}$$

The pass-through matrix Ψ

A pricing model also induces a map from *nominal* marginal costs $\{MC_t\}$ to the price level $\{P_t\}$:

$$P_t = \mathcal{P}_t(\{MC_s\})$$

Log-linearizing around $MC = 1$

$$\hat{P}_t = \sum_{s=0}^{\infty} \frac{\partial \log \mathcal{P}_t}{\partial \log MC_s} \widehat{MC}_s$$

Definition (Pass-through matrix)

$$\Psi_{t,s} \equiv \frac{\partial \log P_t}{\partial \log MC_s}, \qquad \hat{\boldsymbol{P}} = \Psi \cdot \widehat{\boldsymbol{MC}}$$

Interpretation of the columns of Ψ:

- Column s = dynamic price-level response to a one-time shock to nominal MC at date s.
- Row sum $\sum_s \Psi_{t,s}$ = response to a *permanent* shock.
- Long-run neutrality $\Rightarrow \lim_{t\to\infty} \sum_s \Psi_{t,s} = 1$.
- Flexible prices $\Rightarrow \Psi = I$.

$\mathcal{K}$ and Ψ are equivalent representations

Since $\widehat{mc} = \widehat{MC} - \widehat{P}$, plug into $\widehat{P} = \Psi \cdot \widehat{MC}$:

$$\widehat{P} = \Psi\,(\widehat{mc} + \widehat{P}) \quad \Longrightarrow \quad \widehat{P} = \Psi(I - \Psi)^{-1}\,\widehat{mc}$$

Taking first differences ($\pi_t = \widehat{P}_t - \widehat{P}_{t-1}$, i.e. left-multiplying by $I - L$):

$$\boxed{\boldsymbol{\pi} = \underbrace{(I - L)\,\Psi\,(I - \Psi)^{-1}}_{\mathcal{K}}\,\widehat{\boldsymbol{mc}}}$$

- One-to-one mapping between Ψ and $\mathcal{K}$.
- Both encode the full first-order aggregate predictions of the model.
- Direction of inversion: $\Psi = U\mathcal{K}(I + U\mathcal{K})^{-1}$, with $U_{ts} = 1$ for $t \geq s$.

The canonical menu cost model I

Setup (Alvarez–Le Bihan–Lippi 2016; Nakamura–Steinsson 2010):

- Continuum of firms $i \in [0, 1]$, log price p_{it}, optimal log price $p^*_{it} + \log MC_t$
- **Idiosyncratic productivity/demand shocks**: $p^*_{it} = p^*_{it-1} + \epsilon_{it}$, ϵ_{it} symmetric, iid
- Each period a firm pays a quadratic loss $\frac{1}{2}(x_{it} - \log MC_t)^2$ where $x_{it} = p_{it} - p^*_{it}$ is **price gap**
- To change price, pay **menu cost** $\xi_{it} \in \{0, \xi\}$; with probability λ adjustment is **free** ($\xi_{it} = 0$)

$$\min_{x_{it}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2}(x_{it} - \log MC_t)^2 + \xi_{it} \mathbf{1}_{\{x_{it} \neq x_{it-1} - \epsilon_{it}\}} \right] \tag{1}$$

The canonical menu cost model II

Solution to (1) is "*Ss*" rule

$$x_{it} = \begin{cases} x_t^* & \text{w.p. } \lambda \text{ or if } x_{it-1} - \epsilon_{it} \notin [\underline{x}_t, \overline{x}_t] \\ x_{it-1} - \epsilon_{it} & \text{otherwise} \end{cases}$$

$\Rightarrow$ **Two margins** of aggregate adjustment when MC shifts:

- **Extensive margin:** movement of the bands $(\underline{x}_t, \overline{x}_t)$
- **Intensive margin:** movement of the reset point x_t^*

Time-dependent models I

A TD model is governed by a **survival function** Φ_s:

- $\Phi_s \in [0, 1]$, $\Phi_0 = 1$, weakly decreasing in s
- Φ_s is fraction of firms still on their original price s periods after last adjustment
- Adjustment hazard at horizon s: $\lambda_s = \frac{\Phi_{s-1} - \Phi_s}{\Phi_{s-1}}$

Two canonical examples:

Model	Survival function	Hazard
Calvo (1983)	$\Phi_s = (1 - \lambda)^s$	constant λ
Taylor (1979)	$\Phi_s = 1$ for $s < T$, 0 else	spike at $s = T$

Share of price with age s:

$$a_s = \frac{\Phi_s}{\sum_{k=0}^{\infty} \Phi_k}$$

Time-dependent models II

The optimal reset price is given by:

$$x_t^* = arg \min_x \mathbb{E}_0 \sum_{s=0}^{\infty} \beta^s \Phi_s \frac{1}{2} \left(x - \sum_{r=1}^{s} \epsilon_{it+r} - \log MC_{t+s} \right)^2$$

$$x_t^* = \frac{\sum_{s=0}^{\infty} \beta^s \Phi_s \widehat{MC}_{t+s}}{\sum_{s=0}^{\infty} \beta^s \Phi_s} \quad \text{(policy equation)}$$

Price level is weighted average of past reset prices:

$$\hat{P}_t = \sum_{s=0}^{\infty} a_s x_{t-s}^* = \frac{\sum_{s=0}^{\infty} \Phi_s x_{t-s}^*}{\sum_{s=0}^{\infty} \Phi_s} \quad \text{(law of motion)}$$

Pass-through matrix of a TD model

Combining the policy equation with the law of motion we get pass-through matrix:

$$\Psi^{\Phi} = \frac{1}{\sum_{s\geq 0}\Phi_s \; \sum_{s\geq 0}\beta^s\Phi_s} \underbrace{\begin{pmatrix} \Phi_0 & 0 & 0 & \cdots \\ \Phi_1 & \Phi_0 & 0 & \cdots \\ \Phi_2 & \Phi_1 & \Phi_0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}}_{\text{law of motion of prices}} \underbrace{\begin{pmatrix} \Phi_0 & \beta\Phi_1 & \beta^2\Phi_2 & \cdots \\ 0 & \Phi_0 & \beta\Phi_1 & \cdots \\ 0 & 0 & \Phi_0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}}_{\text{response of reset price gap}}$$

Implications:

- Response to a **one-time** shock at $t = 0$: $\quad \widehat{P}_t \propto \Phi_t$.
- Response to a **permanent** shock: $\quad \widehat{P}_t \propto \sum_{s\leq t}\Phi_s$.
- $\Rightarrow$ The survival function Φ can be read off any single impulse response.

Pass-through matrix: Calvo case

For Calvo, $\Phi_s = (1-\lambda)^s$, so

$$\sum_{s\geq 0} \Phi_s = \frac{1}{\lambda}, \qquad \sum_{s\geq 0} \beta^s \Phi_s = \frac{1}{1-\beta(1-\lambda)}$$

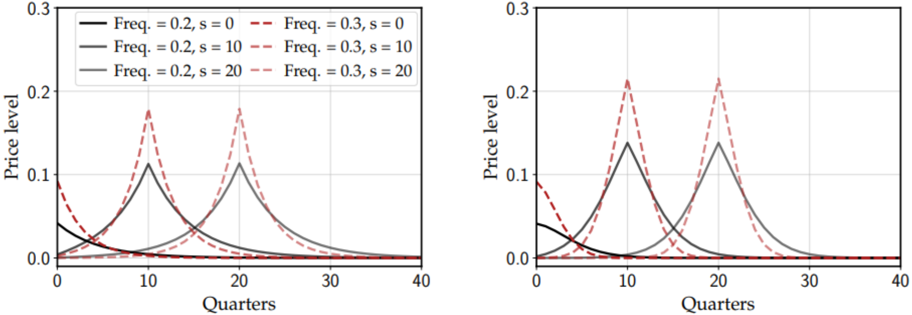
Plugging into Ψ^Φ yields the geometric closed form:

$$\Psi^\Phi = \lambda\big(1-\beta(1-\lambda)\big) \underbrace{\begin{pmatrix} 1 & 0 & 0 & \cdots \\ (1-\lambda) & 1 & 0 & \cdots \\ (1-\lambda)^2 & (1-\lambda) & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}}_{\text{law of motion of prices}} \underbrace{\begin{pmatrix} 1 & \beta(1-\lambda) & \beta^2(1-\lambda)^2 & \cdots \\ 0 & 1 & \beta(1-\lambda) & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}}_{\text{response of reset price gap}}$$

Implications for Calvo:

- Response to a **one-time** shock at $t=0$: $\widehat{P}_t \propto (1-\lambda)^t$ — geometric decay.
- Response to a **permanent** shock: $\widehat{P}_t \propto 1-(1-\lambda)^{t+1}$ — geometric convergence to 1.

Illustration: Calvo



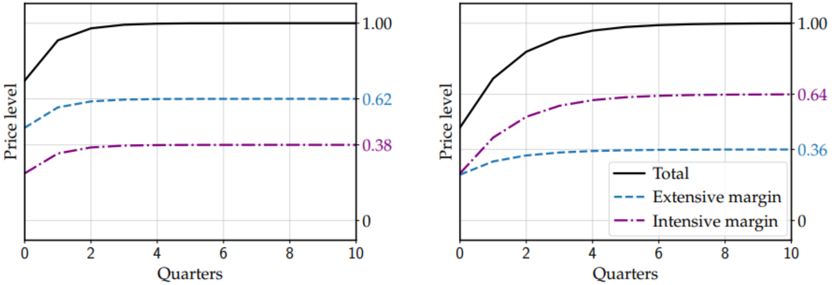
(a) Calvo

(b) Increasing adjustment hazards

Figure 1: Columns $s \in \{0, 10, 20\}$ of TD pass-through matrices

Illustration: SD models

	Golosov-Lucas (GL)	Nakamura-Steinsson (NS)
Menu cost (ξ)	0.0060	0.0513
Prob. of free adj. (λ)	0	0.179
Shock std. (σ_ϵ)	0.046	0.060
Discount factor (β)	0.99	0.99



(a) Golosov-Lucas (b) Nakamura-Steinsson

Main result: Proposition 1

Proposition 1

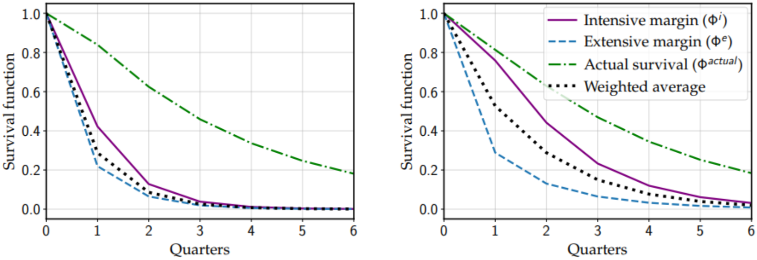
The pass-through matrix of the canonical menu cost model is a **convex combination** of two TD pass-through matrices:

$$\boxed{\Psi \;=\; \alpha\,\Psi^{\Phi^e} \;+\; (1-\alpha)\,\Psi^{\Phi^i}}$$

Define **virtual survival functions** for the two margins:

- $\alpha \sum_{s=0}^{t} \Phi_s^e \big/ \sum_{s\geq 0} \Phi_s^e$ is the impulse response of the price level to a permanent nominal marginal cost shock when only the Ss band shifts;
- $(1-\alpha) \sum_{s=0}^{t} \Phi_s^i \big/ \sum_{s\geq 0} \Phi_s^i$ is the impulse response of the price level to a permanent nominal marginal cost shock when only the reset price gap shifts.

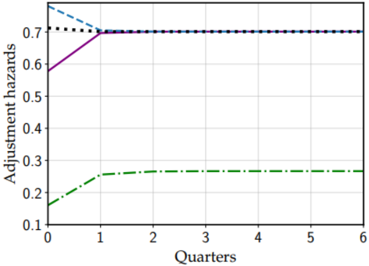
Virtual vs. actual hazards



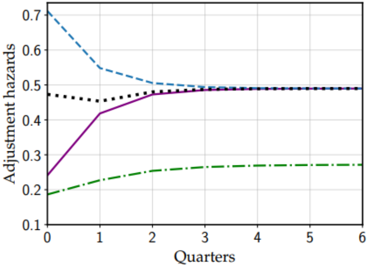
(a) Golosov-Lucas survival functions Φ_t

(b) Nakamura-Steinsson survival functions Φ_t

Numerical equivalence to Calvo



(c) Golosov-Lucas adjustment hazards λ_t



(d) Nakamura-Steinsson adjustment hazards λ_t